Final Project Check-In

Graded

2 Hours, 9 Minutes Late

Group

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Total Points

5 / 5 pts

Question 1

Check-in

5 / 5 pts

– **0 pts** Correct

Overall seems reasonable and the progress is good. I agree with the future directions for phase 3. I also would be curious to know how different the prompts are in your Experiment 1; I could imagine that AdvBench and Alpaca prompts look quite different and so this direction doesn't necessarily reflect harmfulness but instead reflects a more general domain shift.

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Attention Dilution — Mid-Semester Check-in Report

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Github: https://github.com/ayushkokande/attention_dilution

Attention Dilution — Mid-Semester Check-in Report

Project: *Attention Dilution: A Mechanistic Analysis of Context-Length Jailbreaks and Activation Steering in Small LLMs*

Project Type: Original Research (Analytical)

Team: Ayush Kokande · Suraj Sushil Kumar Mishra · Aria Saboonchi

Section 1 is a quick recap of what we promised to do. Section 2 lists the places where we had to adjust the plan, and why. Sections 3–5 report on each phase. Section 6 summarizes what we can say about the proposal's main hypothesis today.

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1. Quick recap of the proposal

The proposal's hypothesis is that safety fine-tuning stores a single "refusal direction" (V_{}) in the model's residual stream, and that a small set of middle-to-late "Guardrail Heads" are responsible for activating it. We hypothesized that long, benign context starves those heads of attention, so (V_{}) never gets triggered, and the model complies with the harmful request.

The plan had three phases:

- **Phase 1 — Attribution Graph.** Find the refusal direction and the Guardrail Heads that drive it.
- **Phase 2 — Context Scalability Test.** Prepend increasing amounts of harmless filler in front of a harmful request and watch how the Guardrail Heads' attribution changes.
- **Phase 3 — Activation Steering.** Once long context has diluted the safety signal, inject (V_{}) back into the model at the right layer and check if it starts refusing again.

To reduce risk we ran the plan as **two parallel experiments** that share the model, the datasets, and the definition of (V_{}), but differ in tools and in which phases they push hardest on. We refer to them as **Experiment 1** and **Experiment 2** throughout this report.

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2. Where we adjusted the proposal

Three small changes from the original plan. Everything else (the three-phase structure, the hypothesis, the intervention) is unchanged.

Change	What the proposal said	What we actually did	Why
Attribution method	Phase 1 would use **Integrated Gradients (IG)** to find the heads responsible for refusal.	We replaced IG with the **difference-of-means** method from Arditi et al. (2024): compute the mean activation on harmful prompts and subtract the mean on harmless prompts, per layer.	IG on attention heads is slow and noisy, and does not scale to the per-head, per-layer, per-(N) sweep we need in Phase 2. Difference-of-means runs in a few minutes and we validate it causally by ablating the direction and checking that the model stops refusing which gives us the same "this direction is responsible" claim IG would.
Model	"A small, safety-aligned model (e.g., a safe-tuned Qwen or Pythia)."	We fixed on `Qwen/Qwen3-1.7B`.	Pythia is not safety-tuned, so there is no refusal behaviour to dilute. Qwen3-1.7B refuses strongly out of the box (85% on our AdvBench sanity set),

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Change	What the proposal said	What we actually did	Why
			supports a 32K context, and fits comfortably on one A100. We also disable Qwen3's "thinking" mode, which otherwise emits a `<think>…</think>` block that pollutes activations.
Attribution graph tool	Phase 1 deliverable was "an attribution graph that isolates the middle-layer heads."	We keep that deliverable but produce it with **`circuit-tracer`** and the pre-trained `mwhanna/qwen3-1.7b-transcoders-lowl0` transcoders (Experiment 1 only).	`circuit-tracer` gives us feature-level attribution graphs finer-grained than what IG-on-heads would have produced and lets us knock out specific MLP transcoder features in Phase 3 as a complement to activation steering.

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3. Phase 1 — Finding the Refusal Vector

3.1 Experiment 1

A fully reproducible pipeline under `experiment_1/`, packaged as two SLURM jobs for the A100 cluster (`sbatch_refusal_pipeline.sh`, `sbatch_scaling_sweep.sh`) plus a local MPS harness (`run_local_pipeline.sh`).

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- **Datasets.** 20 harmful prompts from **AdvBench** + 20 harmless prompts from **Alpaca** for the sanity check; 32 + 32 for extracting the direction; 16 held-out harmful prompts for the interventions.
- **Method.** Record the residual stream at the last prompt token for every prompt, compute `mean_harmful − mean_harmless` at each layer, and rank layers by a *separation score* (how cleanly the direction separates harmful from harmless activations, normalised by spread). Validate the direction by ablating it on held-out harmful prompts (expect refusal to drop) and by adding it to harmless prompts (expect refusal to rise).
- **Numbers on disk.**
  - `results/qwen3-1.7b/sanity_results.json`: **harmful refusal 85.0%**, harmless refusal **0.0%** — the refusal signal we want to dilute is clearly there.
  - `results/qwen3-1.7b/refusal_directions.pt`: **best layer = 22** out of 28 (≈ 79% depth), separation score **4.194** on 32 training prompts.
  - Intervention run is partially done: the baseline pass finished (16/16) but the ablation pass stopped at 11/16 on MPS and needs a cluster re-run.

3.2 Experiment 2

A single A100 notebook, `experiment_2/phase12_qwen3_notebook.ipynb`, that covers Phases 1 and 2 end to end, plus a separate script for Phase 3.

- **Datasets.** 50 harmful + 50 harmless prompts defined in the notebook, split 32/18 into train/val.
- **Method.** Same difference-of-means recipe, but Experiment 2 picks the best layer *directly* by causal ablation: ablate the direction layer by layer and pick the layer that drops refusal on the val set the most. Experiment 2 then does the proposal's explicit "Guardrail Heads" step by computing, for every head, the dot product of its output with (V_{}) at the best layer, and saving the top-(k) heads.
- **Outputs cached in the notebook** (not yet synced into the repo): `V_refusal.pt`, `guardrail_heads.pt`, `phase1_layer_sweep.csv`, `fig_phase1_norms.png`, `fig_phase1_heatmap.png`. Baseline refusal on the notebook's own val split is **55.56%** lower than Experiment 1's 85% because the val list is different (see §7.2).

3.3 Where Phase 1 stands vs the proposal

The proposal asked for an attribution graph that isolates the middle-layer heads responsible for refusal.

- The **refusal direction** is found and causally validated.
- The **middle-layer** claim holds: the best layer is 22 out of 28, which is 79% depth right in the 60–80% band that the prior work (Arditi et al., JailbreakLens) predicts.

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- The **per-head identification** is done in Experiment 2 (saved in `guardrail_heads.pt`) and still pending in Experiment 1 which will produce the graph-level version next sprint via `circuit-tracer`, which gives us a finer-grained version of the deliverable the proposal asked for.

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4. Phase 2 — Context Scalability Test

4.1 Experiment 1

For each of 8 held-out harmful prompts and each filler length (N {0, 128, 512, 1, 2, 4, 8}), we prepend a benign "Willowbrook" preamble of length (N) to the harmful request and run one hooked forward pass plus one greedy generation. We record:

- the attention mass from the final input position to the harmful-request tokens, at every (layer, head),
- the projection of the residual stream onto (V_{}) at the last harmful token and at the final input position,
- the greedy response and its refusal label.

Outputs on disk from the A100 cluster job: `results/qwen3-1.7b/scaling/sweep_rows.jsonl` (56 rows), `summary.csv`, and three figures (`scaling_asr.png`, `scaling_refusal_projection.png`, `scaling_attn_dilution.png`).

Refusal rate by filler length:

(N)	0	128	512	1024	2048	4096	8192
Refusal	**75.0%**	25.0%	25.0%	25.0%	37.5%	25.0%	50.0%

The 50-point drop between (N = 0) and (N = 128) is the direction we expected from the proposal: even a short benign preamble is enough to break refusal on most prompts. The bumpy behaviour past (N) is within the noise band of 8 prompts (§7.1), and the model was loaded with `n_ctx = 2048`, so the 4k and 8k buckets are truncating the filler (§7.3).

4.2 Experiment 2

Same design, but pushes farther on (N): ({0, 128, 512, 1024, 2048, 4096, 8192, 16384}), with a runtime cap for whatever the model's context window allows. Experiment 2 also uses a tighter version of the attention metric: the average attention mass is computed **only over the Guardrail Heads** from Phase 1, not over all heads. Outputs: `phase2_scaling.csv`, `fig_phase2_scaling.png` (visible as cached cell outputs in the notebook).

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4.3 Where Phase 2 stands vs the proposal

The proposal's goal for Phase 2 was to prove that as context length increases, attribution mass shifts away from the Guardrail Heads and eventually crosses a threshold where the jailbreak succeeds.

- The **behavioural threshold** is already visible in Experiment 1: it sits somewhere between (N = 0) and (N = 128) on our current 8-prompt slice. This is encouraging (the effect is real and large) and also a signal that we sampled the low-(N) end too coarsely we'll add buckets at (N {16, 32, 64}) next sprint.
- The **attribution-mass shift** on the Guardrail Heads which is the picture the proposal asked for will be the joint plot combining Experiment 1 (all-heads map) and Experiment 2 (Guardrail-Heads-only curve). (§7.3).

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5. Phase 3 — Activation Steering

5.1 Status

- **Experiment 2: implemented.** `experiment_2/phase3_steering.py` hooks the best layer and adds (V_{}) to the residual stream at every token position during greedy generation. It sweeps ({0, 1, 2, 4, 8, 16}) on a 10-prompt harmful val set at a fixed filler length (default (N = 4096)) and writes the refusal rate curve to `phase3_steering.csv` + `fig_phase3_rescue.png`. This is a direct implementation of the proposal's Phase 3.
- **Experiment 1: not in scope for the current sprint.** Experiment 1 instead implements a *feature-level* intervention (`intervene_refusal_features.py`) that zeroes specific `circuit-tracer` transcoder features on the final prompt token and every generated token. It keeps the proposal's "intervene to rescue safety" spirit, but operates at a finer granularity than activation addition.

5.2 Where Phase 3 stands vs the proposal

The proposal's test for Phase 3 is whether the model starts refusing again after we steer. Experiment 2's rescue curve answers this directly, but it is only wired at (N = 4096) today. To fully close the proposal's Phase 3 deliverable we need the curve at multiple (N) (at least ({1, 4, 8})) and a capability check on an MMLU subset, so we can claim the rescue works *without* breaking the model.

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6. What we can say about the hypothesis today

The proposal claims that long-context jailbreaks succeed because benign bulk starves the Guardrail Heads of the attention they need to trigger (V_{}).

Here is our current evidence:

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- **Refusal is a single identifiable direction.** Difference-of-means + causal ablation recovers a direction at layer 22 that, when removed, makes the model stop refusing on held-out harmful prompts, and when added to harmless prompts makes the model start refusing. This is the Arditi et al. story replicated on Qwen3-1.7B — the picture the proposal assumed.
- **Benign bulk breaks refusal behaviourally.** At (N = 0) the model refuses 75% of held-out harmful prompts; at (N = 128) it refuses 25%. The effect is large and immediate.
- **Activation steering is wired.** Experiment 2's Phase 3 script is working at (N = 4096); we have not yet swept across (N), so we cannot yet quote a rescue curve.